Chapter 12

Chest X-Ray in Aortic Disease

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INTRODUCTION

Today's trainees are raised in a three-dimensional, computed tomographic (CT) radiographic imaging world. Recent trainees tend to "look down" on the chest X-ray for the imaging of the aorta. This chapter is written specifically to bring home the point that important, although at times subtle, clues to the aortic contour are indeed available on the plain chest X-ray [1]. They can be detected by knowing where to look and being interested in looking at the chest X-ray. The information to be garnered can be of vital importance, providing important diagnostic and therapeutic clues even before the CT scan is ordered or performed.

CAPABILITIES OF THE PLAIN CHEST X-RAY

The plain chest X-ray derives its ability to discern the contours of the aorta because of the contrast between the air-filled lungs and the fluid-filled aorta. The normal contours of the aorta are quite familiar and generally very well seen. See Fig. 12.1; it displays a normal chest X-ray as well as one with sites marked where the dilated aorta can make itself known. Also see Fig. 12.2, for example, of thoracic aneurysms diagnosed by chest X-ray.

- The ascending aorta, if substantially enlarged, will appear outside the upper right cardiac silhouette.
- The aortic knob should normally be small and distinct in the upper left mediastinum. This knob will enlarge in the case of arch aneurysm. The knob becomes indistinct in patients with aortic dissection, due to edema and hemorrhage in the tissues.
- The descending aorta normally makes a clean, crisp stripe to the left of the vertebral column. Aneurysmal dilatation of the descending aorta will be clearly seen, with bulging of this stripe to the right.

Also, in cases of severe aortic enlargement, the trachea, left main bronchus, or esophagus may be seen to be displaced on a plain chest X-ray.

Of course, when aortic pathology is suspected on chest radiography, supplementary imaging should be performed using computed tomography angiography (CTA) or magnetic resonance imaging (MRI) [2].

DIAGNOSIS OF ACUTE AORTIC SYNDROME

Chest X-ray can, for the reasons mentioned above, be an important part of the work-up in a patient with acute aortic syndrome. A few specific findings on the chest X-ray may suggest aortic dissection. These include displacement of intimal calcifications, the presence of a double density, or a blurred aortic knob [2,3].

The American Heart Association and American College guidelines for thoracic aortic disease (2010) do recommend the evaluation of mediastinal width in patients being evaluated for aortic dissection if index of suspicion is low [4]. However, it has to be kept in mind that its diagnostic value in aortic dissection is limited and that more than 10% of patients may have a

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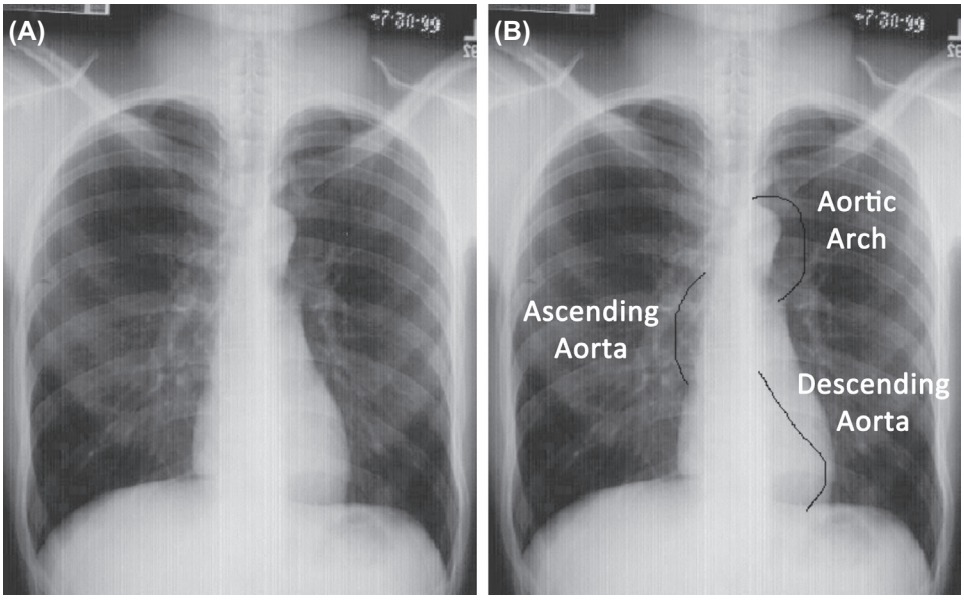


FIGURE 12.1 (A) Normal chest X-ray. (B) Same chest X-ray with areas where ascending, arch, and descending aortic aneurysm shadows may appear.

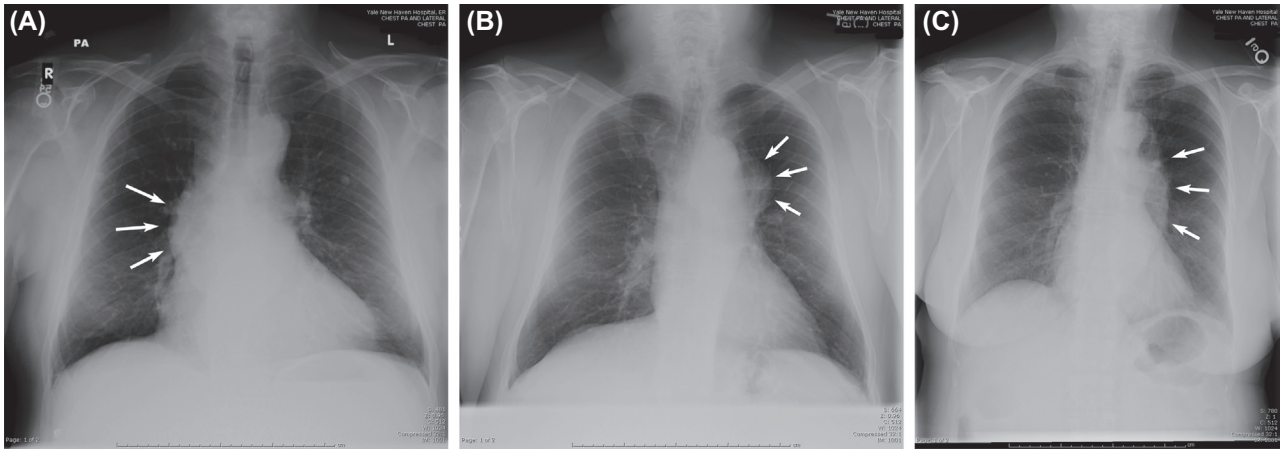


FIGURE 12.2 Aneurysms diagnosed by chest X-ray. (A) Ascending; (B) Arch; (C) Descending.

normal chest X-ray [5,6]. In the International Registry of Aortic Dissection (IRAD) report on the diagnosis of acute aortic dissection, 11.3% and 15.8% of patients had no abnormalities on chest X-ray in acute Type A and B dissection, respectively [6]. Luker et al. reported the evaluation of 75 chest X-rays of patients with aortic dissection and found that only 48% patient chest X-rays evidenced sufficient information to diagnose aortic dissection, exemplifying the known limited sensitivity. In five patients, sufficient findings were overlooked and accurate diagnosis was delayed [5].

Similarly, Hartnell et al. compared chest radiographs of 25 patients with myocardial infarction and 18 patients with acute Type A dissection and came to the conclusion that no individual or combination of radiological signs can accurately discriminate between the two diseases [7].

Somewhat higher accuracy measures were reported by Jagannath et al. and von Kodolitsch et al. [8,9]. Jagannath reviewed chest X-rays of 36 patients with aortic dissection and 36 normal controls. Overall radiologist's impression was a better predictor than individual signs investigated, but interobserver agreement was poor. They found an overall sensitivity of 81% and a specificity of 89% [8]. In the study performed by von Kodolitsch, the authors reevaluated chest X-rays of 216 patients with acute aortic syndrome. They found an overall sensitivity of 64% and a specificity of 86%, with much lower accuracy if the pathology was isolated to the ascending aorta [9].

These findings go in line with a review performed by Klompas, which concluded that chest X-ray is usually abnormal (with a pooled sensitivity of 90%), and that, therefore, a normal chest X-ray substantially lowers the risk of aortic dissection [10].

Mediastinal widening, the most established radiographic sign of aortic dissection, was present in 62.6% of patients with Type A dissection and 56% of patients with Type B dissections in the IRAD report [6]. Pooled sensitivity for mediastinal widening in the review by Klompas was 64% [10].

An abnormal aortic contour, known as a further potential sign of aortic dissection, was present in 46.6% (Type A dissection) and 53% (Type B dissection) of patients in the IRAD study [6]. The review by Klompas reached a pooled sensitivity of 71% for this finding. Displaced intimal calcification is not reliably present in aortic dissection, with a pooled sensitivity of only 9% [10].

CONCLUSIONS

Chest X-ray is a broadly used and easily accessible diagnostic test, which often reveals radiographic signs of aortic disease. Chest X-ray does have an important role in the evaluation of the aorta.

Chest X-ray alone, however, is not a sufficiently accurate diagnostic tool to confirm or exclude aortic pathology, and neither does it offer sufficient anatomic detail to depict accurately the detailed anatomy and complications of aortic aneurysms or allow the planning of therapeutic management. Therefore, the radiographic signs of aortic disease on chest X-ray should lead to further imaging using CTA or MRI. Nevertheless, if a patient is evaluated for aortic dissection and the index of suspicion is high, a CT scan should not be delayed for a chest x-ray.

On the authors' aortic service, we encourage residents trained in computerized tomographic imaging and MRI not to ignore the chest X-ray. The plain film can, upon careful reading, not only raise the suspicion of aortic disease, but also predict, with surprising accuracy, the findings that will obtain on more advanced imaging modalities.

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